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1 Data Scraping Spiders

This section provides an overview of the web scraping spiders used in the TravelHunters project to collect data from various sources.

1.1 Overview of Spiders

Spider Name	Purpose	Source Website	Script Location
booking.py	Scrapes hotel data	Booking.com	scraping_data_files/spiders/booking.py
activities.py	Scrapes activities and tours	GetYourGuide	scraping_data_files/spiders/activities.py
destinations.py	Scrapes destination details	Tourism websites	scraping_data_files/spiders/destinations.py
wikipedia_destinationscraper.py	Scrapes descriptions and descriptions	Wikipedia pages	scraping_data_files/spiders/wikipedia_destinationscraper.py
cityImageScraper.py	Scrapes city images	DuckDuckGo	scraping_data_files/spiders/cityImageScraper.py
googlescraping.py	Scrapes Street View images	Google Maps API	scraping_data_files/spiders/googlescraping.py
selescrape_img.py	Scrapes activity images	GetYourGuide Explorer Pages	scraping_data_files/spiders/selescrape_img.py

1.2 Details of Key Spiders

1.2.1 booking.py

- **Purpose:** Scrapes hotel data from Booking.com, including names, ratings, prices, and locations.
- **Notable Libraries:**
 - Scrapy

- **Key Features:**
 - Handles pagination for worldwide coverage.
 - Extracts structured data such as hotel names, prices, and image URLs.
- **Challenges:**
 - Dynamic content loading handled using Scrapy’s middleware.

1.2.2 activities.py

- **Purpose:** Scrapes activities and tours from GetYourGuide.
- **Notable Libraries:**
 - Scrapy
- **Key Features:**
 - Uses Scrapy to handle static and paginated content.
 - Extracts activity titles, prices, ratings, and image URLs.
- **Challenges:**
 - Handling rate-limiting and ensuring efficient scraping of large datasets.

1.2.3 wikipedia_destination_scraper.py

- **Purpose:** Scrapes city descriptions and images from Wikipedia pages.
- **Notable Libraries:**
 - hashlib
 - urllib
 - beautiful soup
 - wikipedia_destination_links
- **Key Features:**
 - Extracts main images, coordinates, and descriptions for cities.
 - Handles multilingual pages where applicable.
- **Challenges:**
 - Ensuring data consistency across different Wikipedia page structures.

1.2.4 destinations.py

- **Purpose:** Scrapes destination details from various tourism websites.
- **Notable Libraries:**
 - Scrapy
- **Key Features:**
 - Extracts destination names, descriptions, and geographical coordinates.
 - Handles multiple website structures using Scrapy's flexible selectors.
- **Challenges:**
 - Ensuring consistency across data from different sources.

1.2.5 cityImageScraper.py

- **Purpose:** Scrapes city images using DuckDuckGo search.
- **Notable Libraries:**
 - duckduckgo_search
- **Key Features:**
 - Retrieves image URLs for cities based on search queries.
 - Filters and validates image links to ensure quality.
- **Challenges:**
 - Handling rate-limiting and ensuring relevant search results.

1.2.6 googlescraping.py

- **Purpose:** Scrapes Street View images of hotels and destinations using the Google Maps API.
- **Notable Libraries:**
 - Google Maps API client
- **Key Features:**
 - Retrieves Street View images based on latitude and longitude.
 - Extracts metadata such as coordinates and image timestamps.
- **Challenges:**
 - Managing API quotas and ensuring accurate geolocation data.

1.2.7 geoscraper.py

- **Purpose:** Sub-script of googlescraping.py. Create perimeter of coordinates for google-scraping.py
- **Key Features:**
 - Takes coordinates of each city and creates a perimeter of coordinates based of city center
 - formats them into a json
- **Challenges:** assuring quality in image sets

1.2.8 selescrape_img.py

- **Purpose:** Scrapes activity images from GetYourGuide Explorer pages.
 - **Notable Libraries:**
 - Selenium
 - **Key Features:**
 - Extracts image URLs for activities and tours.
 - Handles pagination and structured data extraction.
 - downloads jpg from image URLs, renames with city name, and stores in folder
 - **Challenges:**
 - Bypassing getyourguide anti scraping features
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